

DP Computer Science: A 1.2.3 Logic Gates

1. Core Concepts & Learning Objectives

Learning Intentions

By the end of this lesson, students will be able to:

- 1. State what a **logic gate** is and what it does to its binary inputs
- 2. Describe the **purpose** of logic gates inside a computer system, and their **role in binary computing**
- 3. Describe the **behaviour** of each of the seven Boolean operators (**AND, OR, NOT, NAND, NOR, XOR, XNOR**) in terms of the output produced for its inputs

Command Term Note: All three intentions sit at **Describe** — a detailed account. Constructing and analysing truth tables belongs to A1.2.4 and is deliberately not an intention here.

Key Vocabulary (Tier 3)

Term	Definition
Logic gate	An electronic component that takes one or two binary inputs and produces one binary output by a fixed rule
Boolean operator	The logical rule a gate applies, such as AND, OR, or NOT
Inverter	Another name for a NOT gate, because it flips its single input
Truth table	A table listing every possible input combination and the output for each. Used here to read gate behaviour; constructing them is A1.2.4
Exclusive	In XOR and XNOR, "exclusive" means the "or both" case is treated differently from an ordinary OR

Pedagogical Focus

- **Communication Skills (ATL):** Students consciously practise stating a gate's behaviour precisely. The Describe command term rewards exact phrasing over gist. Name the skill at the point of the oracy task and reflect on it in the exit check.
- **Differentiation (ATT):** This is a term-dense Describe lesson—seven gate names, their operators, and the vocabulary of inputs, outputs, and inversion, all new at once. Differentiation leads because the barrier here is **language, not concept**.

Name the skill to students: "*Today you are practising precise description—not just naming a gate, but describing what it does.*"

EAL Scaffold

Support Type	Description
Sentence Frame	"A [gate name] gate produces an output of 1 when ____ ; otherwise, it outputs 0."
Vocabulary Strategy	Example and non-example cards to keep on the desk
Input Skeleton	00, 01, 10, 11 provided so effort goes on reading outputs

2. Lesson Sequence (~45 Minutes)

Part 1: Do Now (Retrieval) — ~10%

Activity: Three short recall items:

Question	Source
1. Write the denary number 11 in 4-bit binary	[A1.2.1]
2. How many bits are in one byte, and roughly how many characters can one byte encode?	[A1.2.2]
3. In binary, what do 0 and 1 stand for physically inside a circuit?	[A1.2.2] / [A1.1]

Part 2: Main Sequence — ~60%

Phase 1: Pre-Teach the Vocabulary (Example and Non-Example) (~8 mins)

Step 1 — Umbrella Term: Logic Gate (Fraye Model)

Definition	Characteristics
An electronic component that takes one or two binary inputs and produces one binary output by a fixed rule	Takes inputs; applies a rule; produces one output
Example	Non-Example
An AND gate that outputs 1 only when both inputs are 1	A transistor (it amplifies signals; it doesn't apply a Boolean rule)

Step 2 — Confusable Pairs

Introduce each gate against the gate it is most often mistaken for:

Pair	Why They Are Confusable
OR vs. XOR	Both output 1 when at least one input is 1; XOR excludes the "both 1" case
AND vs. NAND	NAND is the inverse of AND (outputs are opposite)
XOR vs. XNOR	XNOR is the inverse of XOR (outputs are opposite)

Phase 2: Teacher Exposition — Concrete to General (~15 mins)

Step 1 — Start with Three Basic Gates

Gate	Symbol	Rule (Sentence Frame)	Truth Table
AND	&	"Outputs 1 only when both inputs are 1"	00→0, 01→0, 10→0, 11→1
OR	≥1	"Outputs 1 when either input is 1 (including both)"	00→0, 01→1, 10→1, 11→1
NOT	¬	"Outputs the opposite of its single input"	0→1, 1→0

Step 2 — Build the Derived Gates (Inversions)

Gate	Relationship	Rule
NAND	NOT + AND	"Outputs 0 only when both inputs are 1; otherwise, 1"
NOR	NOT + OR	"Outputs 1 only when neither input is 1; otherwise, 0"

Step 3 — The Exclusive Pair

Gate	Rule	Contrast with
XOR	"Outputs 1 when exactly one input is 1"	OR includes the "both 1" case
XNOR	"Outputs 1 when both inputs are the same "	XOR outputs 1 when inputs differ

Phase 3: Oracy and Production Task (Communication) (~7 mins)

Activity (Pairs):

1. **Speak:** Students take turns describing a named gate to a partner using the sentence frame
2. **Check:** Partner checks against the master table
3. **Write:** Each student writes a one-line precise description for the two gates they find hardest to separate (e.g., XOR and XNOR)

Mini-whiteboards make the descriptions visible for quick correction.

Part 3: Deliberate Practice — ~20%

Activity: A1.2.3 Worksheet, completed individually:

Task Type	Description
State Output Rule	For each gate, state the rule in words
Read Single-Gate Outputs	Given inputs, determine output (including XNOR)
Purpose and Use	Short questions on why gates are used in computer systems

 This is Describe-level practice: identify and state behaviour, not evaluate multi-gate circuits.

Optional Extension: Logic Gate Adventure Game — The Boolean Caverns (recall extension)

Part 4: Exit Check — ~10%

Two Items (Readable in Minutes):

Question	Purpose
Part A: "Using the sentence frame, describe what a NAND gate does to its inputs."	Checks understanding of derived gate behaviour
Part B: "In one sentence, describe the difference between XOR and XNOR."	Doubles as the Communication reflection: precise description, not identification

Expected Answers:

Part A	"A NAND gate produces an output of 0 only when both inputs are 1; otherwise, it outputs 1."
Part B	"XOR outputs 1 when exactly one input is 1; XNOR outputs 1 when both inputs are the same (0,0 or 1,1)."

3. Complete Reference Table: Seven Logic Gates

Gate	Symbol	Rule	Truth Table
AND	&	Outputs 1 only when both inputs are 1	00→0, 01→0, 10→0, 11→1
OR	≥1	Outputs 1 when either input is 1 (including both)	00→0, 01→1, 10→1, 11→1
NOT	¬	Outputs the opposite of its single input	0→1, 1→0
NAND	↑	Outputs 0 only when both inputs are 1; otherwise 1	00→1, 01→1, 10→1, 11→0
NOR	↓	Outputs 1 only when neither input is 1; otherwise 0	00→1, 01→0, 10→0, 11→0
XOR	⊕	Outputs 1 when exactly one input is 1	00→0, 01→1, 10→1, 11→0
XNOR	⊙	Outputs 1 when both inputs are the same	00→1, 01→0, 10→0, 11→1

Confusable Pairs — Side-by-Side Comparison

Pair	Gate A	Rule A	Gate B	Rule B	Key Difference
OR vs. XOR	OR	1 if either input is 1 (11→1)	XOR	1 if exactly one input is 1 (11→0)	XOR excludes the "both 1" case
AND vs. NAND	AND	1 if both inputs are 1	NAND	0 if both inputs are 1	NAND is the inverse of AND
XOR vs. XNOR	XOR	1 if inputs differ	XNOR	1 if inputs are the same	XNOR is the inverse of XOR

4. Why Logic Gates Matter in Binary Computing

Purpose	Description
Decision Making	Gates allow computers to make decisions based on binary inputs (e.g., if X AND Y are true, do Z)
Arithmetic	Combinations of gates form adders , which perform addition and other arithmetic
Memory	Gates can be arranged into flip-flops , which store binary data (the basis of memory)
Control	Gates decode instructions and generate control signals inside the CPU

Purpose	Description
Binary Operations	All computation is ultimately a series of logic gate operations

"Logic gates are the components that let a computer **act** on binary signals rather than only store or move them."

5. Differentiation & Extension

Support (Removable Scaffolding)

Support Strategy	Description
Example/Non-Example Cards	Vocabulary cards to keep on the desk
Input Skeleton	00, 01, 10, 11 provided; effort goes on reading outputs
Sentence Frame	Visible during oracy task; removed for exit check
Master Table	Complete truth table available for reference

Extension (Deeper Rather Than More)

Challenge Strategy	Description
Why NAND Alone?	Describe, in words only, why NAND alone could stand in for NOT, AND, and OR — a purpose-and-use observation
Purpose Reasoning	Explain why understanding gate behaviour is essential for understanding how computers work

Hold this at description. Building or analysing NAND-only circuits is beyond A1.2.3 and is not set here.

6. Example Hinge Questions

Question	Purpose
"What is a logic gate, and what does it do to its inputs?"	Checks understanding of the umbrella term
"How does an AND gate differ from a NAND gate?"	Checks understanding of inversion
"What is the key difference between OR and XOR?"	Checks understanding of the "exclusive" concept
"What does the NOT gate do to its input?"	Checks understanding of inversion
"How are logic gates used inside a computer system?"	Checks understanding of purpose and role

7. Learning Objectives (Summary)

Knowledge and Understanding

- Explain the purpose and function of logic gates in digital circuits
- Identify and describe the behaviour of AND, OR, NOT, NAND, NOR, XOR, and XNOR gates
- Understand the relationship between Boolean operators and corresponding logic gates

ATL Skills

Skill	Description
Communication	Explain the function of logic gates precisely
Thinking	Apply logical reasoning to understand gate behaviour
Self-Management	Organise and complete vocabulary and description tasks

8. Resources

- Presentation (Slide Deck)
- InThinking eBook (A1.2.3 with quiz)
- Worksheet: Gate descriptions + purpose questions
- Worksheet Answers
- Logic Gate Adventure Game: The Boolean Caverns (optional recall extension)

Author's Reflection

"This lesson was already an exposition, and it worked, so the job here was to tighten it rather than rebuild it. The main change is scope. The previous version reached into constructing and analysing truth tables and reading circuits, which belongs to A1.2.4 and A1.2.5, and trying to carry all three in one place taught the same ideas thinly and three times over. Holding this lesson to Describe—the behaviour of each gate and what gates are for—keeps the load down and leaves construction where it is properly taught.

The real barrier at Describe is the vocabulary—seven gate names and their operators arriving at once—so I pre-teach the terms by example and non-example and introduce each confusable gate against its neighbour, then have students state each gate in their own precise words. I also reworked the worksheet: XNOR was missing, the Boolean-expression questions overshot Describe and have gone, and short purpose-and-use questions now cover the parts of the statement the old sheet left out. Done well, this lesson gives the next two the vocabulary and the gate behaviour they rely on."